

NATIONAL CURRICULUM EXAMINATION

Mathematics - Senior 3 (S3)

Unit 1: Algebraic Fractions

Time: 1 Hour 30 Minutes

Total Marks: 100

SECTION A: SHORT ANSWER QUESTIONS (50 Marks)

1. Simplify the following algebraic fraction to its lowest terms: (5 marks)

$$\frac{12x^2y}{18xy^2}$$

2. Find the values of x for which the following fraction is undefined: (5 marks)

$$\frac{x + 5}{x^2 - 9}$$

3. Simplify by factorizing the numerator and denominator: (5 marks)

$$\frac{x^2 - 5x + 6}{x - 2}$$

4. Perform the operation and simplify: (5 marks)

$$\frac{3}{a} + \frac{2}{b}$$

5. Multiply the following fractions: (5 marks)

$$\frac{4t^2}{3} \times \frac{9}{2t}$$

6. Divide the following and simplify the result: (5 marks)

$$\frac{x + 1}{5} \div \frac{2x + 2}{10}$$

7. Express as a single fraction in its simplest form: (5 marks)

$$\frac{2}{x-1} - \frac{3}{x+1}$$

8. Simplify the expression:

(5 marks)

$$\frac{x^2 - y^2}{x + y}$$

9. Solve for x:

(5 marks)

$$\frac{2x}{3} = 4$$

10. State the LCM of the denominators in the following expression:

(5 marks)

$$\frac{1}{x} + \frac{1}{2x} + \frac{1}{x^2}$$

SECTION B: STRUCTURED QUESTIONS (50 Marks)

11. (a) Simplify the following expression completely:

(8 marks)

$$\frac{2x^2 - 8}{x^2 + 4x + 4}$$

(b) Hence or otherwise, evaluate the expression when $x = 3$.

(4 marks)

12. Solve the following algebraic equation for x:

(12 marks)

$$\frac{3}{x-2} + \frac{1}{x+2} = \frac{4}{x^2 - 4}$$

13. A rectangle has an area of $\frac{x^2 - 16}{x + 2}$ square units and a width of $(x - 4)$ units.

(a) Find an expression for the length of the rectangle in terms of x.

(6 marks)

(b) Simplify the expression for the length as much as possible.

(6 marks)

14. Worker A takes x hours to complete a task, and Worker B takes $(x + 2)$ hours.

(a) Write an expression for the fraction of the task they complete together in one hour.

(6 marks)

(b) If they complete $\frac{5}{12}$ of the task in one hour together, find the value of x.

(8 marks)